



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

UNIT TEST

07-08-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8
	A.	B,C,D	A,B,C	A,B,C,D	A,C,D	B,D	A,C	A,B,D	A,B,D
SECTION-II	Q.	1	2	3	4	5	6		
	A.	2.00	1.62	3.14	10.00	2.03	3.33		
SECTION-III	Q.	1	2	3	4	5	6		
	A.	5	2	4	8	2	0		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8
	A.	A,C,D	A,B,C	B,C,D	A,C,D	A,C,D	A,C	B,C	A,B,C
SECTION-II	Q.	1	2	3	4	5	6		
	A.	2.00	5.18	4.00	2.50	3.00	2.00		
SECTION-III	Q.	1	2	3	4	5	6		
	A.	8	7	6	6	6	8		

PART-3 : MATHEMATICS

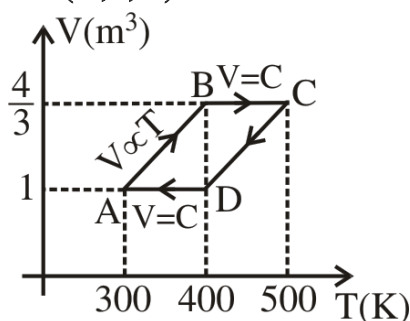
SECTION-I	Q.	1	2	3	4	5	6	7	8
	A.	C	A,B,D	A,C	A,C	A,B	A,C,D	A,B,C	A,B
SECTION-II	Q.	1	2	3	4	5	6		
	A.	0.50	4.00	49.50	0.35 or 0.36	1009.00	6054.00		
SECTION-III	Q.	1	2	3	4	5	6		
	A.	1	2	6	5	3	1		

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (B,C,D)

Line CD does not pass oxygen. So, it's not $V \propto T$.

$$W_{DA} = W_{BC} = 0$$

$$W_{DA} = nC_p \Delta T = 2 \left[\frac{7}{2} R \right] [400 - 300] = 700R$$

Process equ. of CD,

$$V - 1 = \frac{1}{300} (T - 400)$$

$$\Rightarrow V = \frac{T}{300} - \frac{1}{3} = \frac{PV}{600R} - \frac{1}{3}$$

$$P = \frac{600R}{V} \left(V + \frac{1}{3} \right) = 600R \left(1 + \frac{1}{3V} \right)$$

$$W = \int_{4/3}^1 P dV = 600R \left(V + \frac{1}{3} \ln V \right)_{4/3}^1$$

$$= 600R \left[\frac{-1}{3} + \frac{1}{3} \ln \left(\frac{3}{4} \right) \right]$$

$$W = -200R \left(1 + \ln \frac{4}{3} \right)$$

$$W_{\text{Cycle}} = 700R - 200R - 200R \ln \left(\frac{4}{3} \right)$$

$$= 500R - 200R \ln \left(\frac{4}{3} \right)$$

$$= 100R \left(5 - \ln \frac{16}{9} \right)$$

As process is reversible $\Rightarrow \Delta S_{\text{universe}} = 0$

As process is cyclic $\Rightarrow \Delta S_{\text{system}} = 0$

$$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surrounding}}$$

$$0 = 0 + \Delta S_{\text{surrounding}}$$

$$\Rightarrow \Delta S_{\text{surrounding}} = 0$$

$$\Delta S_{\text{CD}} = nR \ln \frac{V_2}{V_1} + nC_V \ln \frac{T_2}{T_1}$$

$$= 2R \ln \left(\frac{3}{4} \right) + 2 \left(\frac{5}{2} R \right) \ln \left(\frac{4}{5} \right)$$

$$= R \left[\ln \left(\frac{9}{16} \right) + \ln \left(\frac{4}{5} \right)^5 \right] = R \ln \left[\frac{(3)^2}{(5)^5} \times (4)^3 \right]$$

3. Ans (A,B,C,D)

$$\phi = \frac{IR}{L} = \frac{\left(\frac{100V}{100\Omega} \right) (50\Omega)}{10m} = \frac{5V}{m}$$

$$\phi(2) = \epsilon_1 \Rightarrow \epsilon_1 = 10V$$

$$\phi(3) = \epsilon_2 \Rightarrow \epsilon_2 = 15V$$

$$\phi(2.5) = \frac{r_1 r_2}{r_1 + r_2} \left(\frac{10}{r_1} + \frac{15}{r_2} \right) = 12.5$$

$$\frac{10}{r_1} + \frac{15}{r_2} = \frac{12.5}{r_1} + \frac{12.5}{r_2}$$

$$\frac{2.5}{r_2} = \frac{2.5}{r_1} \Rightarrow r_1 = r_2$$

4. Ans (A,C,D)

$$\leftarrow \text{O} \rightarrow \frac{qQ}{4\pi\epsilon_0 r^2}$$

$$(A) \frac{mv^2}{r} + \frac{qQ}{4\pi\epsilon_0 r^2} = qvB$$

$$v^2 - \frac{qBr}{m} v + \frac{qQ}{4\pi\epsilon_0 rm} = 0$$

$$v^2 - (10^{-2})(100)(10)v + \frac{9 \times 10^9 (10^{-2})(10^{-6})}{10} = 0$$

$$v^2 - 10v + 9 = 0$$

$$v = 1 \text{ m/s or } 9 \text{ m/s}$$

$$(B) qvB = \frac{qQ}{4\pi\epsilon_0 r^2} \Rightarrow v(100) = \frac{9 \times 10^9 \times 10^{-6}}{100}$$

$$v = 0.9 \text{ m/s}$$

$$\text{but as } r \uparrow \frac{qQ}{4\pi\epsilon_0 r^2} < qvB \Rightarrow \text{particle will curve}$$

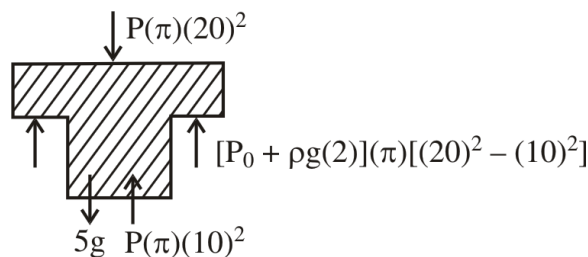
$$(C) \frac{1}{2} m(v^2) + \frac{qQ}{4\pi\epsilon_0 r_1} = \frac{qQ}{4\pi\epsilon_0 r_2} + \frac{1}{2} m(v_f^2)$$

$$v_f^2 = v^2 + \frac{2qQ}{4\pi\epsilon_0 m} \left[\frac{1}{r_1} - \frac{1}{r_2} \right]$$

$$= 25 + 2 \times 9 \times 10^9 \times 10^{-2} \times 10^{-6} \left(\frac{1}{10} - \frac{1}{5} \right)$$

$$v_f = \sqrt{25 - 18} = \sqrt{7} \text{ m/s}$$

5. Ans (B,D)



$$P(\pi)(10^{-2}) + P_0(\pi)(0.2)^2 + \rho g \left(\frac{2}{100} \right) (\pi)(0.2)^2$$

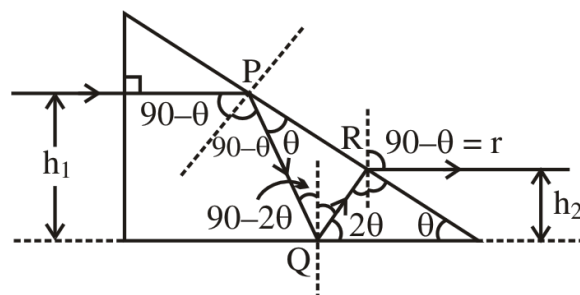
$$= P_0(\pi)(0.2)^2 + 5g + \left[P_0 + \rho g \frac{(2)}{100} \right] (10^{-2})\pi$$

$$P = P_0 + \rho g \left(\frac{2}{100} \right) [-3] + \frac{5g}{\pi \times 10^{-2}}$$

$$= 10^5 - 10^3 \times 10 \times \frac{6}{100} + \frac{5000}{\pi}$$

$$= 10^5 - 600 + \frac{5000}{\pi}$$

6. Ans (A,C)



$$i + 90 + 3\theta = 180 \Rightarrow i = 90 - 3\theta$$

$$\mu \sin i = 1 \sin r$$

$$\mu \sin (90 - 3\theta) = 1 \sin (90 - \theta)$$

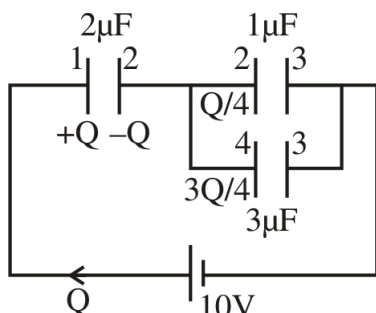
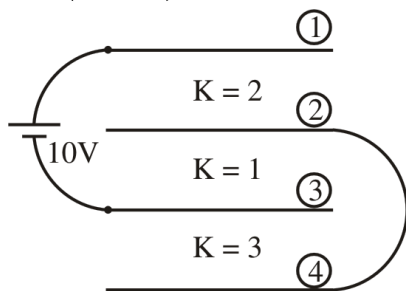
$$\mu = \frac{\cos \theta}{\cos 3\theta}$$

$$h_1 = PQ \cos (90 - 2\theta) = PQ \sin 2\theta$$

$$h_2 = QR \cos (90 - 2\theta) = QR \sin 2\theta$$

$$\frac{h_1}{h_2} = \frac{PQ}{QR} = \frac{\sin[3\theta]}{\sin \theta}$$

8. Ans (A,B,D)



$$C_{eq} = \frac{2 \times 4}{2 + 4} \mu F = \frac{4}{3} \mu F$$

$$Q_2 = -Q + \frac{Q}{4} = \frac{-3Q}{4} = \frac{-3}{4} \left(\frac{4}{3} \mu F \right) (10V) = -10V$$

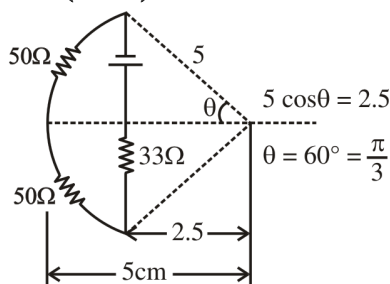
$$Q_3 = -Q = - \left(\frac{4}{3} \mu F \right) (10V) = -\frac{40}{3} \mu C$$

$$U = \frac{1}{2} C_{eq} V^2 = \frac{1}{2} \left(\frac{4}{3} \mu F \right) (100 V^2) = \frac{200}{3} \mu J$$

PART-1 : PHYSICS

SECTION-II

1. Ans (2.00)



$$\epsilon_{in} = \frac{d\Phi_B}{dt} = A \frac{dB}{dt} = \left\{ \frac{\pi}{3} (5)^2 - \left(\frac{5}{2} \right)^2 \sqrt{3} \right\} \frac{dB}{dt}$$

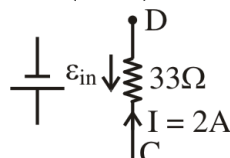
$$= 25 \left\{ \frac{\pi}{3} - \frac{\sqrt{3}}{4} \right\} \{4\pi + 3\sqrt{3}\} \text{ volt}$$

$$= \frac{25}{12} [(4\pi - 3\sqrt{3}) (4\pi + 3\sqrt{3})]$$

$$= \frac{25}{12} [160 - 27] = \frac{25}{12} \times 133$$

$$I = \frac{\epsilon_{in} - \epsilon}{R_{eq}} = \frac{\left(\frac{25}{12} - \frac{1}{2} \right) (133)}{133} = 2A$$

2. Ans (1.62)



$$\epsilon_{in} = \left[\frac{d\Phi_B}{dt} \right] = A \frac{dB}{dt}$$

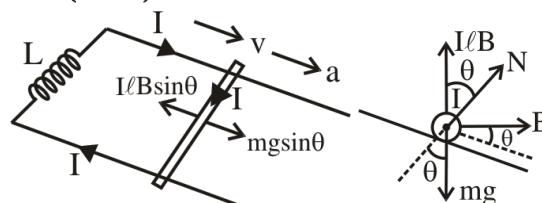
$$= \left[\frac{1}{2} \left(\frac{5}{2} \right)^2 \sqrt{3} \right] [4\pi + 3\sqrt{3}]$$

$$= \frac{25}{8} \sqrt{3} [4\pi + 3\sqrt{3}] \text{ volt}$$

$$V_C - V_D = +2(33) + \frac{25}{8} [4\pi\sqrt{3} + 9]$$

$$x = 162.87 \text{ volt}$$

3. Ans (3.14)



$$mg \sin \theta - I \ell B \sin \theta = m \frac{dv}{dt} \quad \dots(1)$$

$$v \ell B \sin \theta - L \frac{dI}{dt} = 0 \quad \dots(2)$$

$$mg \sin \theta - I \ell B \sin \theta = m \left[\frac{L}{\ell B \sin \theta} \frac{d^2 I}{dt^2} \right]$$

$$\frac{d^2 I}{dt^2} = - \frac{\ell B \sin \theta}{mL} \left\{ \ell B \sin \theta \left(I - \frac{mg}{\ell B} \right) \right\}$$

$$I = \frac{mg}{\ell B} + I_0 \sin \left\{ \frac{\ell B \sin \theta}{\sqrt{mL}} t + \phi \right\}$$

$$\text{at } t = 0, I = 0$$

$$I = \frac{mg}{\ell B} \left\{ 1 - \cos \left(\frac{\ell B \sin \theta}{\sqrt{mL}} t \right) \right\}$$

$$v = \frac{L}{\ell B \sin \theta} \frac{dI}{dt} = \frac{L}{\ell B \sin \theta} \left\{ \frac{mg}{\ell B} \left(0 + \sin \left(\frac{\ell B \sin \theta}{\sqrt{mL}} t \right) \frac{\ell B \sin \theta}{\sqrt{mL}} \right) \right\}$$

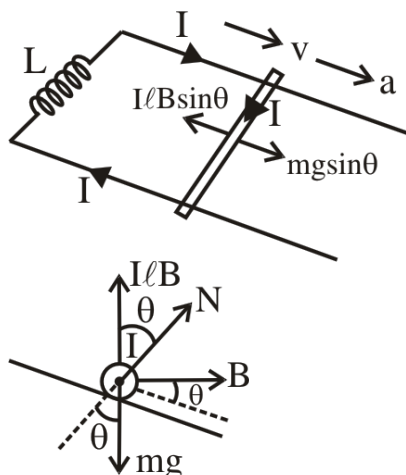
$$v = \frac{\sqrt{mL} g}{\ell B} \sin \left(\frac{\ell B \sin \theta}{\sqrt{mL}} t \right)$$

$$\text{to lose contact } I = \frac{mg}{\ell B}$$

$$\Rightarrow t = \frac{\pi \sqrt{mL}}{2 \ell B \sin \theta} = \frac{\pi \sqrt{2 \times 2}}{2 \times 1 \times 2 \times \sin 30^\circ}$$

$$t = \pi = 3.14 \text{ s}$$

4. Ans (10.00)



$$mg \sin \theta - I l B \sin \theta = m \frac{dv}{dt} \quad \dots(1)$$

$$v l B \sin \theta - L \frac{dI}{dt} = 0 \quad \dots(2)$$

$$mg \sin \theta - I l B \sin \theta = m \left[\frac{L}{l B \sin \theta} \frac{d^2 I}{dt^2} \right]$$

$$\frac{d^2 I}{dt^2} = - \frac{l B \sin \theta}{m L} \left\{ l B \sin \theta \left(I - \frac{mg}{l B} \right) \right\}$$

$$I = \frac{mg}{l B} + I_0 \sin \left\{ \frac{l B \sin \theta}{\sqrt{m L}} t + \phi \right\}$$

$$\text{at } t = 0, I = 0$$

$$I = \frac{mg}{l B} \left\{ 1 - \cos \left(\frac{l B \sin \theta}{\sqrt{m L}} t \right) \right\}$$

$$v = \frac{L}{l B \sin \theta} \frac{dI}{dt}$$

$$= \frac{L}{l B \sin \theta} \left\{ \frac{mg}{l B} \left(0 + \sin \left(\frac{l B \sin \theta}{\sqrt{m L}} t \right) \frac{l B \sin \theta}{\sqrt{m L}} \right) \right\}$$

$$v = \frac{\sqrt{m L} g}{l B} \sin \left(\frac{l B \sin \theta}{\sqrt{m L}} t \right)$$

$$v = \frac{\sqrt{2 \times 2}}{1 \times 2} (10) = 10 \text{ m/s}$$

5. Ans (2.03)

$$F_g = F_b + T$$

$$\sigma \left[\pi R^2 (3R) - \frac{2}{3} \pi R^3 \right] g$$

$$= \rho \left[\pi R^2 (3R) - \frac{2}{3} \pi R^3 \right] g + T$$

$$T = \rho \left[\frac{7}{3} \pi R^3 \right] g$$

6. Ans (3.33)

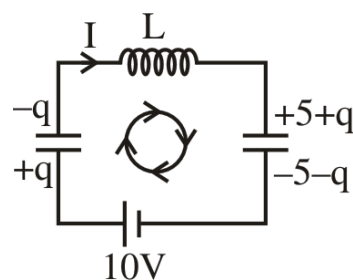
$$F_B = \rho \left[\pi R^2 (3R) - \frac{2}{3} \pi R^3 \right] g = F_2 - \rho g R (\pi R^2)$$

$$F_2 = \frac{7}{3} \pi R^3 \rho g + \pi R^3 \rho g = \frac{10}{3} \pi R^3 \rho g$$

PART-1 : PHYSICS

SECTION-III

1. Ans (5)



$$10 - \frac{q}{1} - L \frac{dI}{dt} - \frac{5+q}{1} = 0$$

When $I \rightarrow \max$

$$\frac{dI}{dt} = 0$$

$$10 - \frac{q}{1} - 5 \frac{q}{1} = 0$$

$$q = 2.5 \mu\text{C}$$

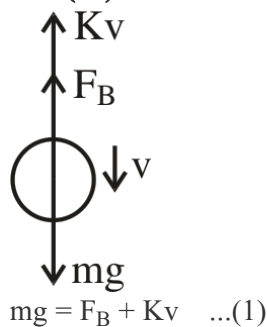
$$\frac{(5)^2}{2C_2} + q\epsilon = \frac{q^2}{2C_1} + \frac{1}{2} L I^2 + \frac{(q+5)^2}{2C_2}$$

$$I = \sqrt{\frac{1}{L} \left[2q\epsilon - \frac{q^2}{C_1} - \frac{(q+5)^2}{C_2} + \frac{(5)^2}{C_2} \right]}$$

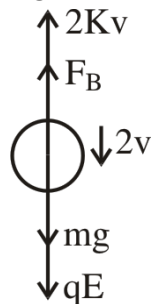
$$I = \sqrt{2[2 \times 25 - 6.25 - 56.25 + 25] \times 10^{-6}}$$

$$I = \sqrt{25 \times 10^{-6}} \text{ A} = 5 \text{ mA}$$

2. Ans (2)

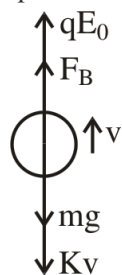


$$mg = F_B + Kv \quad \dots(1)$$



$$qE + mg = F_B + 2Kv \quad \dots(2)$$

$$qE = Kv$$



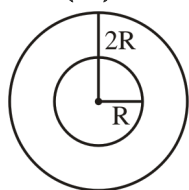
$$qE_0 + F_B = mg + Kv$$

$$qE_0 + F_B = F_B + Kv + Kv$$

$$qE_0 = 2qE$$

$$E_0 = 2E$$

3. Ans (4)



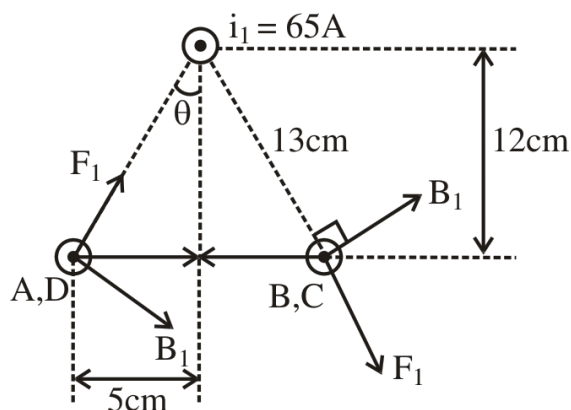
$$\begin{cases} 0 & ; r < R \\ \frac{\mu_0 I}{2\pi r} & ; R \leq r \leq 2R \\ 0 & ; 2R < r \end{cases}$$

$$U = \frac{1}{2} LI^2 = \int_R^{2R} \frac{1}{2\mu_0} \left(\frac{\mu_0 I}{2\pi r} \right)^2 (2\pi r dr \ell)$$

$$\frac{1}{2} LI^2 = \frac{\mu_0 I^2 \ell}{4\pi} \int_R^{2R} \frac{dr}{r} = \frac{\mu_0 I^2 \ell}{4\pi} \ln(2)$$

$$L = \frac{\mu_0 \ell}{2\pi} \ln(2)$$

4. Ans (8)



$$\tau = (F_1 \cos \theta)(AB)$$

$$= [I_2 (AD) B_1 \sin 90^\circ] \left[\frac{12}{13} \right] \left[\frac{10}{100} m \right]$$

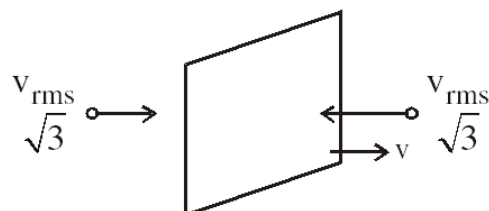
$$= (78) \left(\frac{10}{100} \right) \left(\frac{2 \times 10^{-7} \times 65}{0.13} \right) \left(\frac{12}{13} \right) (0.1)$$

$$= \frac{78 \times 65 \times 24}{169} \times 10^{-7} Nm$$

$$= 6 \times 5 \times 24 \times 10^{-7} Nm$$

$$\frac{x}{90} = \frac{30 \times 24}{90} = 8$$

5. Ans (2)



In Δt time interval,

$$F_{drag} = F_{front} - F_{back}$$

$$= \rho \left[A \left(v + \frac{v_{rms}}{\sqrt{3}} \right)^2 \right] - \rho A \left[v - \frac{v_{rms}}{\sqrt{3}} \right]^2$$

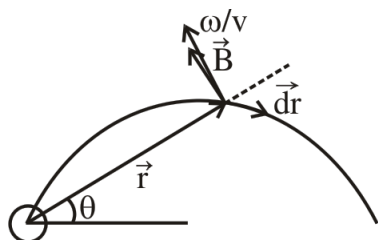
$$= \rho A \left[\frac{4}{\sqrt{3}} v_{rms} v \right] = 4\rho A v \sqrt{\frac{RT}{M_w}}$$

$$F_{drag} = 4\rho A \sqrt{\frac{RT}{M_w}} v_0 = 4 \times 2 \times 10^{-4} \sqrt{\frac{25}{3} \times 300}$$

$$= 4 \times 10^{-3} \sqrt{25 \times 10^4}$$

$$= 4 \times 10^{-3} \times 5 \times 10^2 = 2N$$

6. Ans (0)



$$B = \frac{\mu_0 I}{2\pi r} \quad \dots(1)$$

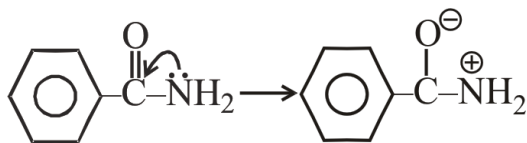
As angle between $\vec{B}$ and $\vec{V}$ is zero induced emf will also be zero.

PART-2 : CHEMISTRY

SECTION-I

5. Ans (A,C,D)

'B' does not give this reaction



Only 1° amine give this reaction

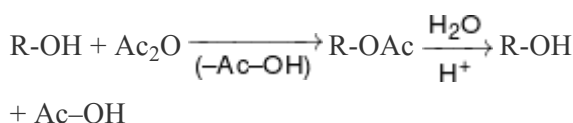
PART-2 : CHEMISTRY

SECTION-II

1. Ans (2.00)

cd

5. Ans (3.00)



let member of hydroxyl group in alcohol be x which are acylated further.

Moles of KOH required per mole of acylated alcohol to hydrolyse = x

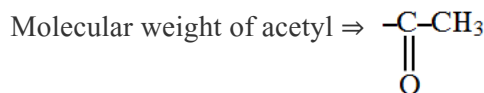
$$x \left[\frac{0.436}{92 - x + 43x} \right] = \frac{0.336}{56}$$

$$x = 3$$

6. Ans (2.00)

Difference in molecular weight = 190 - 106 = 84

$$\text{Given } \begin{cases} \text{Molecular weight of } C_4H_{10}O_3 = 12 \times 4 + 10 + 16 \times 3 \\ 190 = 48 + 10 + 48 \\ = 106 \end{cases}$$



$$\Rightarrow 12 \times 2 + 3 + 16 \Rightarrow 43$$

Late know that one hydrogen replace by one acetal group.

Then number of acetyl is 2

PART-2 : CHEMISTRY

SECTION-III

1. Ans (8)

$$x = 6$$

$$y = 1$$

$$z = 3$$

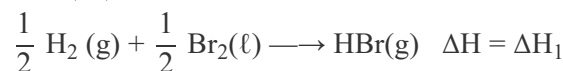
2. Ans (7)

$$PT = \text{constant} \setminus PV^{1/2} = \text{constant}$$

$$\therefore C = C_v + R/1-n = 3R/2 + 2R = 7R/2$$

$$\therefore q = nCDT = 2 \times 7R/2 (300 - 600) = -2100R$$

3. Ans (6)



At constant pressure (Reversible process)

$$q_p = \Delta H_1$$

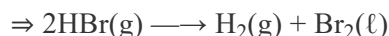
$$\therefore \Delta S_{\text{sys}} + \Delta S_{\text{surr}} = 0$$

$$\therefore \Delta S_{\text{sys}} = -170 \text{ J/K}$$

$$\therefore \Delta S_{\text{sys}} = \frac{\Delta H_1}{T} \text{ (Reversible)}$$

$$\therefore \Delta H_1 = 300 \times (-170) \text{ J}$$

$$\Delta H_1 = -51000 \text{ J}$$



$$\Delta H = -2 \Delta H_1$$

$$\Delta H = -2 \times (-51000) \text{ J}$$

At constant volume ($q_v = \Delta U$)

$$\therefore \Delta U = \Delta H - \Delta n_g RT$$

$$= (2 \times 51000) - [(-1) \times 8.3 \times 300]$$

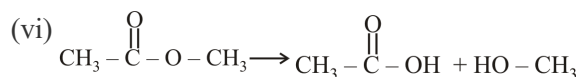
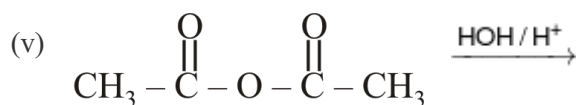
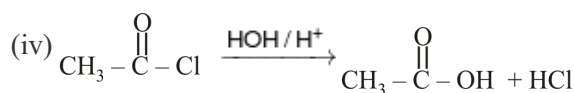
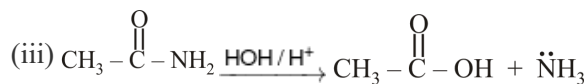
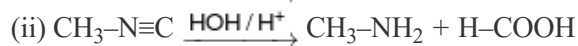
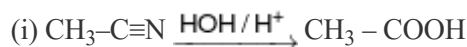
$$= 104490 \text{ J}$$

$$\therefore \Delta S = \frac{q_v}{T} = \frac{104490}{300} = 348.3$$

$$\therefore \Delta S = 348.3 \text{ J/K}$$

5. Ans (6)

Compound which will give carboxylic acid on acidic hydrolysis are (i), (ii), (iii), (iv), (v) & (vi)



6. Ans (8)

$x = 3$

$y = 5$

PART-3 : MATHEMATICS

SECTION-I

1. Ans (C)

$$\int \frac{1}{\sqrt{2} + \sqrt{2}\sin(x + \frac{\pi}{4})} dx = \frac{1}{\sqrt{2}} \int \frac{1 - \sin(x + \frac{\pi}{4})}{\cos^2(x + \frac{\pi}{4})} dx$$

$$\Rightarrow \frac{1}{\sqrt{2}} \left[\int \sec^2(x + \frac{\pi}{4}) - \int \sec(x + \frac{\pi}{4}) \tan(x + \frac{\pi}{4}) dx \right]$$

$$\Rightarrow \frac{1}{\sqrt{2}} \left[\tan(x + \frac{\pi}{4}) - \sec(x + \frac{\pi}{4}) \right] + C$$

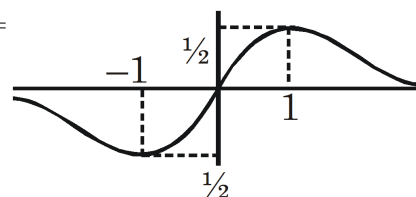
Equation of curve should have two independent arbitrary constants.

2. Ans (A,B,D)

$d((1+x^2)y) = dx$

$(1+x^2)y =$

$x + C$



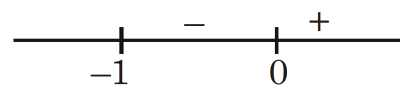
$\Rightarrow y = \frac{x}{1+x^2} \quad \{\because f(0) = 0\}$

3. Ans (A,C)

(A) Let $\alpha > 1$ and $f(x) = (1+x)^\alpha - (1+\alpha x)$

if $x > -1$

$\Rightarrow f'(x) = \alpha(1+x)^{\alpha-1} - \alpha$

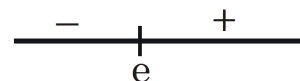


$\Rightarrow f(x) \geq f(0) = 0$

$\Rightarrow (1+x)^\alpha > 1 + \alpha x$

(C) Let $f(x) = x - e \ln x$

$f'(x) = \frac{x-e}{x}$



$\Rightarrow f(x) \geq f(e) = 0$

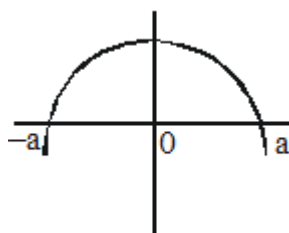
$\Rightarrow x \geq e \ln x$

4. Ans (A,C)

$ay = 3(a-x)(a+x)$

Area = $2 \int_0^a \frac{3}{a} (a^2 - x^2) dx = 64$

$\Rightarrow \frac{3}{a} \left(\left(a^2 x - \frac{x^3}{3} \right) \right)_0^a = 32$



$\Rightarrow \frac{3}{a} \left(a^3 - \frac{a^3}{3} \right) = 32$

$\Rightarrow 3 \left(a^2 - \frac{a^2}{3} \right) = 32 \Rightarrow 3a^2 - a^2 = 32$

$\Rightarrow 2a^2 = 32 \Rightarrow a^2 = 16 \Rightarrow a = \pm 4$

5. Ans (A,B)

$$A = 2 \int_0^1 \frac{dx}{1+x^4}$$

$$x \in (0, 1) \Rightarrow 1+x^2 > 1+x^4 \Rightarrow \frac{1}{1+x^4} > \frac{1}{1+x^2}$$

$$A = 2 \int_0^1 \frac{dx}{1+x^4} > 2 \int_0^1 \frac{dx}{1+x^2} > \frac{\pi}{2}$$

$$\text{similarly } 1+x > 1+x^4 \Rightarrow 2 \int_0^1 \frac{dx}{1+x^4} > 2 \ln 2$$

6. Ans (A,C,D)

Let P(x, y), equation of normal at P

$$\Rightarrow Y - y = \frac{-1}{\left(\frac{dy}{dx}\right)}(X - x)$$

$$A \left(x + y \frac{dy}{dx}, 0 \right) \quad B \left(0, y + \frac{x}{\left(\frac{dy}{dx}\right)} \right)$$

$$\frac{BP}{PA} = \frac{1}{2} \Rightarrow y \frac{dy}{dx} = 2x \Rightarrow y^2 = 2x^2 + C$$

∵ Curve passes through (0, 4)

$$\text{So } y^2 = 2x^2 + 16$$

Now check options

7. Ans (A,B,C)

$$S = 4 \left[\int_0^{\pi/8} e^{\cos 4x} dx + \int_{\pi/8}^{\pi/4} e^{-\cos 4x} dx \right]$$

Put (4x) = t,

$$S = 2 \int_0^{\pi/2} e^{\sin t} dt$$

$$\text{Also, } 2 \int_0^{\pi/2} 1 dt < 2 \int_0^{\pi/2} e^{\cos t} dt$$

$$< 2 \int_0^{\pi/2} e^t dt$$

$$\Rightarrow \pi < S < 2(e^{\pi/2} - 1)$$

8. Ans (A,B)

$$x dy = y dx + \sin\left(\frac{1}{x}\right) dx$$

$$\frac{dy}{dx} = \frac{y}{x} + \frac{1}{x} \sin\left(\frac{1}{x}\right)$$

$$P = -\frac{1}{x}; \text{ I.F.} = e^{\log\left(\frac{1}{x}\right)} = \frac{1}{x}$$

$$\frac{y}{x} = \int \frac{1}{x^2} \sin\left(\frac{1}{x}\right) dx$$

$$= - \int \sin t dt = \cos \frac{1}{x} + c$$

$$y = x \left(\cos \frac{1}{x} + c \right)$$

$$y = x \left(\cos \frac{1}{x} + 3 \right) \text{ is on such}$$

$$y = x \left(\cos \frac{1}{x} + 1 + 2 \right) = x \left(2 \cos^2 \frac{1}{2x} + 2 \right)$$

$$= 2x \left(2 \cos^2 \frac{1}{2x} + \sin^2 \frac{1}{2x} \right)$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (0.50)

$$\frac{dy}{dx} - y = e^x g'(x)$$

$$\Rightarrow y e^{-x} = \int g'(x) dx$$

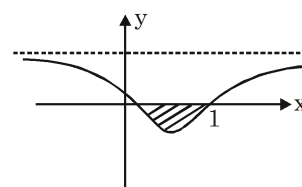
$$\Rightarrow y e^{-x} = \frac{x(x-1)}{x^2+1} e^{-x} + c$$

$$y = \frac{x(x-1)}{x^2+1} + c e^x \Rightarrow c = 0 \quad \because f(0) = 0$$

$$\Rightarrow f(x) = \frac{x(x-1)}{(x^2+1)}$$

$$\frac{dy}{dx} = \frac{x^2+2x-1}{(x^2+1)^2} \Rightarrow y'(1) = \frac{1}{2}$$

$$\text{Area} \int_0^1 \frac{x^2-x}{x^2+1} dx$$



$$\int_0^1 \left(1 - \frac{x}{x^2+1} - \frac{1}{x^2+1} \right) dx$$

$$= 1 - \frac{1}{2} \ln 2 - \frac{\pi}{4} = \ln \left(\frac{e}{\sqrt{2}} \right) - \frac{\pi}{4}$$

2. **Ans (4.00)**

$$\frac{dy}{dx} - y = e^x g'(x)$$

$$\Rightarrow ye^{-x} = \int g'(x) dx$$

$$\Rightarrow ye^{-x} = \frac{x(x-1)}{x^2+1} e^{-x} + c$$

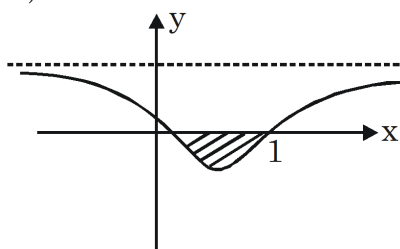
$$y = \frac{x(x-1)}{x^2+1} + ce^x \Rightarrow c = 0 \quad \because f(0) = 0$$

$$\Rightarrow f(x) = \frac{x(x-1)}{(x^2+1)}$$

$$\frac{dy}{dx} = \frac{x^2+2x-1}{(x^2+1)^2} \Rightarrow y'(1) = \frac{1}{2}$$

Area

$$\int_0^1 \frac{x^2-x}{x^2+1} dx$$



$$\int_0^1 \left(1 - \frac{x}{x^2+1} - \frac{1}{x^2+1} \right) dx$$

$$= 1 - \frac{1}{2} \ln 2 - \frac{\pi}{4} = \ln \left(\frac{e}{\sqrt{2}} \right) - \frac{\pi}{4}$$

3. **Ans (49.50)**

$$f(x) + f(1-x) = \frac{1+3x^2-3x}{3} + \frac{-1+2x-2x^2}{2} + 1 + \frac{1}{6}$$

$$= \frac{2+6x^2-6x-3+6x-6x^2}{6} + 1 + \frac{1}{6}$$

$$= -\frac{1}{6} + 1 + \frac{1}{6}$$

$$\therefore f(x) + f(1-x) = 1$$

$$\therefore f\left(\frac{1}{100}\right) + f\left(\frac{2}{100}\right) + \dots + f\left(\frac{99}{100}\right)$$

$$= f\left(\frac{1}{100}\right) + f\left(\frac{99}{100}\right) + \dots + f\left(\frac{49}{100}\right) + f\left(\frac{51}{100}\right) + f\left(\frac{1}{2}\right)$$

$$= 1 \times 49 + \frac{1}{24} - \frac{1}{8} + \frac{1}{2} + \frac{1}{12} = \frac{99}{2}$$

4. **Ans (0.35 or 0.36)**

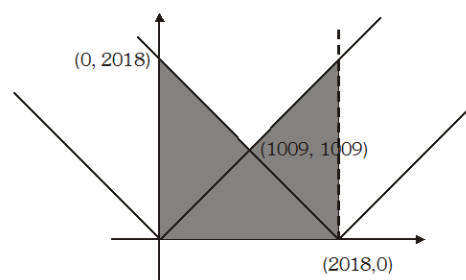
$$I = \int_{1/7}^{6/7} f(f(x)) dx = \int_{1/7}^{6/7} f(f(1-x)) dx$$

$$I = \int_{1/7}^{6/7} f(1-f(x)) = \int_{1/7}^{6/7} 1 - f(f(x)) dx$$

$$\therefore 2I = \frac{5}{7}$$

$$\Rightarrow I = \frac{5}{14}$$

5. **Ans (1009.00)**



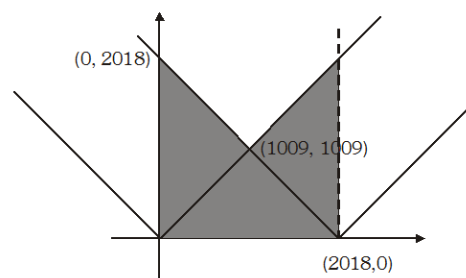
$$\int_0^{2018} f(x) dx = \text{shaded area} = 3 \left(\frac{1}{2} (2018)(1009) \right)$$

$$= 3 (1009)^2$$

$$\text{similarly } \int_0^{2018} g(x) dx = 2018 \int_0^1 g(x) dx$$

$$= \frac{2018}{4} = \frac{1009}{2}$$

6. **Ans (6054.00)**



$$\int_0^{2018} f(x) dx = \text{shaded area} = 3 \left(\frac{1}{2} (2018)(1009) \right)$$

$$= 3 (1009)^2$$

$$\text{similarly } \int_0^{2018} g(x) dx = 2018 \int_0^1 g(x) dx$$

$$= \frac{2018}{4} = \frac{1009}{2}$$

PART-3 : MATHEMATICS

SECTION-III

1. **Ans (1)**

By A.M - G.M.

$$\sqrt[3]{f(x) \cdot 1 \cdot 1} \leq \frac{f(x) + 1 + 1}{3} = \frac{f(x) + 2}{3}$$

$$\Rightarrow \int_0^1 \sqrt[3]{f(x)} dx \leq \frac{1}{3} \int_0^1 f(x) dx + \int_0^1 \frac{2}{3} dx = \frac{1}{3} + \frac{2}{3} = 1$$

Similarly

$$\int_0^1 \sqrt[5]{f(x)} dx \leq \frac{1}{5} \int_0^1 f(x) dx + \int_0^1 \frac{4}{5} dx = \frac{1}{5} + \frac{4}{5} = 1$$

$$\text{and } \int_0^1 \sqrt[7]{f(x)} dx \leq 1$$

Multiplying, we get

$$\left(\int_0^1 \sqrt[3]{f(x)} dx \right) \cdot \left(\int_0^1 \sqrt[5]{f(x)} dx \right) \cdot \left(\int_0^1 \sqrt[7]{f(x)} dx \right) < 1$$

$$\Rightarrow I \leq 1$$

2. **Ans (2)**

$$I = \int_0^1 \frac{x^2 - 2x + 2 + x^2 - 1}{\sqrt{x^2 - 2x + 2}} dx$$

$$= \int_0^1 \left(\sqrt{x^2 - 2x + 2} + \frac{(x-1)}{\sqrt{x^2 - 2x + 2}} \cdot (x+1) \right) dx$$

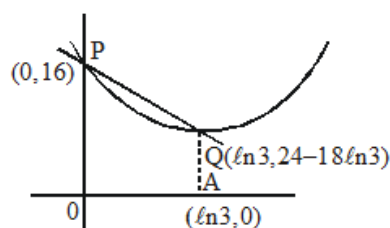
$$= \int_0^1 \left(\frac{d}{dx} (x+1) \sqrt{x^2 - 2x + 2} \right) dx = 2 - \sqrt{2}$$

3. **Ans (6)**

$$\text{Let } f(x) = \sqrt{1 + (x-1)^3}$$

$$\Rightarrow f^{-1}(x) = 1 + (x^2 - 1)^{1/3}$$

4. **Ans (5)**



$$\text{Area of trap OAQP} = \frac{1}{2} (40 - 18\ln 3) \ln 3$$

Req. area

$$= \frac{1}{2} (40 - 18\ln 3) \ln 3 - \int_0^{\ln 3} (e^{2x} - 18x + 15) dx$$

$$= 5\ln 3 - 4$$

5. **Ans (3)**

$$a = 20 - 6\pi, b = 10\pi - 30, c = 10 - 3\pi$$

$$\Rightarrow a + b + c = \pi$$

$$\therefore \alpha = -1 \text{ and } \beta = 1$$

$$\Rightarrow \tan^{-1} \alpha + \cos^{-1} (\beta - 1) = -\frac{\pi}{4} + \frac{\pi}{2} = \frac{\pi}{4}$$

6. **Ans (1)**

Given,

$$2xy \frac{dy}{dx} = x^2 + y^2 \Rightarrow \frac{2xy dy - y^2 dx}{x^2} = dx$$

$$\Rightarrow d \left(\frac{y^2}{x} \right) = dx$$

Integrating, $\frac{y^2}{x} = x + c$, since it passes through (2, 1)

$$\Rightarrow c = -\frac{3}{2}$$

$$\therefore y^2 = x^2 - \frac{3}{2}x \Rightarrow \left(x - \frac{3}{4} \right)^2 - y^2 = \frac{9}{16}$$

Clearly hyperbola is rectangular, so its eccentricity $e = \sqrt{2}$

$$\therefore e^2 - 1 = 1$$